Introduction to Machine Learning · Fall 2026 Focus College · Instructor: Arpan Bhavsar

Assignment 2 — Model Bake-Off: Comparing Supervised Models

Weight	**5%** of the course grade (100 points)
Released	Week 4, Saturday, October 24, 2026
Due	**Saturday, October 31, 2026, before midnight Pacific Time** (the day of the Week 5 class)
Work	Individual, on your team's dataset (teammates share the data, not the code or the write-up)
Submit	`A2_LastnameFirstname.ipynb` + `A2_LastnameFirstname_Report.pdf` (2–3 pages) on the LMS
Outcomes	LO2, LO4, LO6, LO7, LO8, LO9

> **Download:** this brief plus `assignment2-assets.zip` (a runnable starter notebook, the report template and a submission checklist). **Head start:** Week 4 lab Parts 5 to 7 are a complete worked example of this assignment on EarlyCheck; the starter notebook reproduces them before you switch to your own data.

1. The question

Assignment 1 built a clean, leak-free pipeline and a first classifier for your team's problem. Assignment 2 asks: **which supervised model should your project actually use, and why?** You will run a fair competition, tune the strongest contenders, look at their errors, and defend a champion with evidence. As in class, "best" will rarely be decided by one number.

2. Tasks

1. **Models in one pipeline.** Train at least six models in the same pipeline: Dummy, logistic regression, KNN, Naive Bayes, SVM and random forest (a single decision tree is a bonus). Reuse your Assignment 1 preprocessing; every model gets exactly the same steps inside `make_pipeline` / `ColumnTransformer`. Choose the Naive Bayes variant that fits your features (GaussianNB for numeric tables).
2. **Fair comparison.** Compare all models with stratified 5-fold cross-validation (one `StratifiedKFold` object reused by every model) using your chosen metric plus accuracy; report mean ± standard deviation in a table. Say which differences are larger than about one standard deviation.

3. **Tuning.** Tune at least two models with GridSearchCV or RandomizedSearchCV on the training data only; report the search ranges, the best hyperparameters and the CV score before and after.
4. **Visual evaluation.** Open the test set once, after all choices are made: confusion matrix and ROC curves for your top three models, and feature importance for the random forest.
5. **Champion.** Choose and justify a champion: strengths, weaknesses, what the errors look like (who is missed, who is falsely flagged), and one way to reduce its biggest weakness.

Deliverables: a notebook that runs top to bottom (restart and run all before submitting) and a 2–3 page report with the comparison table (use `report-template.md`).

3. Marking rubric (100 points → 5%)

Criterion	Points	Excellent (90–100%)	Needs work (<60%)
All required models implemented correctly in pipeline	25	Correct, complete, justified with evidence	Missing, incorrect or unexplained
Cross-validation and comparison table	20	Correct, complete, justified with evidence	Missing, incorrect or unexplained
Hyperparameter tuning	15	Correct, complete, justified with evidence	Missing, incorrect or unexplained
Visual evaluation (confusion matrix, ROC, importance)	15	Correct, complete, justified with evidence	Missing, incorrect or unexplained
Champion justification and interpretation	15	Correct, complete, justified with evidence	Missing, incorrect or unexplained
Code quality and report clarity	10	Correct, complete, justified with evidence	Missing, incorrect or unexplained
Total	**100**		

4. Rules that cost marks if broken

> - **No tuning on the test set.** Tune with cross-validation on training data; use the test set once (remember Baidu's 200 ImageNet submissions and the peeking simulation).
> - **No causal claims from importances.** Write "the model relies on X to rank risk", not "X causes" or "reducing X will prevent" (the asthma trap, Caruana et al. 2015).
> - **No leakage.** Scaling, imputation and encoding are fitted inside the pipeline, per fold.

> - **Large data:** SVM and kNN are slow on hundreds of thousands of rows. A stratified sample (for example 20,000 rows) is fine; say so in the report.

5. Policies

- **Late work:** zero unless an extension is approved at least 24 hours before the deadline.
- **AI tools:** allowed for learning and debugging, with an APA-style citation and a short note on how they were used. You must be able to explain every line you submit; I may ask you to walk through your code.
- **Academic integrity:** code and write-up are individual even though teammates share a dataset.

Questions: abhavsar@focuscollege.ca, or ask in class. EarlyCheck and all course health examples are learning exercises, not medical advice.